9) $y'' + 3y' + 3.25y = 3\cos(t) - 1.5\sin(t)$

$$\lambda^2 + 3\lambda + 3.25 = 0$$

$$\lambda = \frac{-3 \pm \sqrt{3^2 - 4(1)(3.25)}}{2(1)}$$

$$\lambda = \frac{-3 \pm \sqrt{-4}}{2}$$

$$\lambda = -1.5 \pm i$$

Complex roots $\Rightarrow$ $y_h(t) = e^{-1.5t}(A\cos(t) + B\sin(t))$

$r(t) = 3\cos(t) - 1.5\sin(t) \rightarrow y_p(t) = K\cos(t) + M\sin(t)$

$$y_p'(t) = -K\sin(t) + M\cos(t)$$
$$y_p''(t) = -K\cos(t) - M\sin(t)$$

$$[-K\cos(t) - M\sin(t)] + 3[-K\sin(t) + M\cos(t)] +$$
$$3.25[K\cos(t) + M\sin(t)] =$$

$$-K\cos(t) - M\sin(t) - 3K\sin(t) + 3M\cos(t) + 3.25K\cos(t) +$$
$$3.25M\sin(t) =$$

$$(-K + 3M + 3.25K)\cos(t) + (-M - 3K + 3.25M)\sin(t) =$$
$$(3M + 2.25K)\cos(t) + (-3K + 2.25M)\sin(t) \Rightarrow$$

$$3M + 2.25K = 3 \quad \Rightarrow M = 0.4,\ K = 0.8$$
$$-3K + 2.25M = -1.5$$

$$\boxed{y(t) = e^{-1.5(t)}(A\cos(t) + B\sin(t)) + 0.8\cos(t) + 0.4\sin(t)}$$

11)

$$4y'' - 15y' - 4y = 0$$

$$y'' - 15/4\, y' - y = 0 =$$

$$y_1 = y$$

$$\boxed{\begin{aligned} y_1' &= y_2 \\ y_2' &= 15/4\, y_2 + y_1 \end{aligned}}$$

$$y' = Ay = \begin{bmatrix} 0 & 1 \\ 1 & 15/4 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{bmatrix} -\lambda & 1 \\ 1 & 15/4 - \lambda \end{bmatrix} = 0$$

$$-\lambda(15/4 - \lambda) - 1 = 0$$

$$-15/4\,\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - 15/4\,\lambda - 1 = 0$$

$$\lambda = 4,\ \lambda = -1/4$$

$\lambda = 4$

$$(A - 4I)x = 0 \Rightarrow \left[\begin{array}{cc|c} -4 & 1 & 0 \\ 1 & 15/4 - 16/4 & 0 \end{array}\right] \Longrightarrow$$

$$= \left[\begin{array}{cc|c} -4 & 1 & 0 \\ 1 & -1/4 & 0 \end{array}\right]$$

$$\left.\begin{aligned} -4x_1 + x_2 &= 0 \\ 1x_1 - 1/4\, x_2 &= 0 \end{aligned}\right\} x_2 = 4x_1$$

$$x^{(1)} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

3) $\lambda = 2$

$(A - 2I)x = 0$

$$\begin{bmatrix} -1 & 2 \\ 1/2 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} -x_1 + 2x_2 = 0 \\ 1/2\,x_1 - x_2 = 0 \end{matrix} \Rightarrow x_1 = 2x_2$$

$$x = \begin{bmatrix} 2c \\ c \end{bmatrix} \Rightarrow x^{(1)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$\lambda = 0, \quad (A - 0I)x = 0$

$$\begin{bmatrix} 1 & 2 \\ 1/2 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x_1 + 2x_2 = 0 \\ 1/2\,x_1 + x_2 = 0 \end{matrix} \Rightarrow x_1 = -2x_2$$

$$x^{(2)} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$y = c_1\begin{bmatrix} 2 \\ 1 \end{bmatrix}e^{2t} + c_2\begin{bmatrix} 2 \\ -1 \end{bmatrix}e^{0t}$$

$$= c_1\begin{bmatrix} 2 \\ 1 \end{bmatrix}e^{2t} + c_2\begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\boxed{y_1 = 2c_1e^{2t} + 2c_2 \qquad y_2 = c_1e^{2t} - c_2}$$

11) $\lambda = -1/4$

$$(A + 1/4 I)x = 0 \Rightarrow \left[\begin{array}{cc|c} 1/4 & 1 & 0 \\ 1 & 4 & 0 \end{array}\right] \Rightarrow \begin{array}{l} 1/4 x_1 + x_2 = 0 \\ x_1 + 4x_2 = 0 \end{array} \Rightarrow x_2 = -1/4 x_1$$

$$x^{(2)} = \begin{bmatrix} 1 \\ -1/4 \end{bmatrix}$$

$$\boxed{y(t) = c_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{-4t} + c_2 \begin{bmatrix} 1 \\ -1/4 \end{bmatrix} e^{-1/4 t}}$$

3)

$$y_1' = y_1 + 2y_2$$

$$y_2' = \frac{1}{2} y_1 + y_2$$

$$y' = Ay = \begin{bmatrix} 1 & 2 \\ 1/2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 2 \\ 1/2 & 1-\lambda \end{bmatrix} = (1-\lambda)^2 - 1 \Rightarrow$$

$$(1-\lambda)(1-\lambda) - 1 = 0$$

$$1 - \lambda - \lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - 2\lambda = 0$$

$$\lambda = 2, 0$$

11) $y_1' = 2y_1 + 5y_2$
$y_2' = -\frac{1}{2}y_1 - \frac{3}{2}y_2$ $\quad y_1(0) = 12 \quad y_2(0) = 0$

$$y' = Ay = \begin{bmatrix} 2 & 5 \\ -1/2 & -3/2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\det(A - \lambda I) = \det\left(\begin{bmatrix} 2-\lambda & 5 \\ -1/2 & -3/2-\lambda \end{bmatrix}\right) =$$

$$(2-\lambda)(-3/2-\lambda) - 5/2$$

$$-3 - 2\lambda + 3/2\lambda + \lambda^2 - 5/2$$
$$\lambda^2 - 1/2\lambda - 1/2 = 0$$
$$\lambda = 1, -1/2$$

$\lambda = 1, (A - I)x = 0$

$$\begin{bmatrix} 1 & 5 \\ -1/2 & -5/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \qquad x_1 + 5x_2 = 0$$
$$x_1 = -5x_2$$
$$x^{(1)} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

$\lambda = -1/2, (A + 1/2 I)x = 0$

$$\begin{bmatrix} 5/2 & 5 \\ -1/2 & -4/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \qquad 5/2 x_1 + 5x_2 = 0$$
$$x_1 = -2x_2 \qquad x^{(2)} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

11)

$$Y = C_1\begin{bmatrix}5\\-1\end{bmatrix}e^{t} + C_2\begin{bmatrix}2\\-1\end{bmatrix}e^{-1/2t}$$

$$y_1 = 5c_1e^{t} + 2c_2e^{-1/2t} \qquad y_2 = -c_1e^{t} - c_2e^{-1/2t}$$

$$y_1(0) = 5c_1 + 2c_2 = -12 \qquad y_2(0) = -c_1 - c_2 = 0$$

$$-c_1 = c_2$$

$$5c_1 - 2c_1 = -12$$

$$3c_1 = -12$$

$$c_1 = -4,\ c_2 = 4$$

$$\boxed{y_1 = -20e^{t} + 8e^{-1/2t} \qquad y_2 = 4e^{t} - 4e^{-1/2t}}$$

9) $y_1' = 4y_1 + y_2$

$y_2' = 4y_1 + 4y_2$

$$Y' = AY = \begin{bmatrix}4 & 1\\ 4 & 4\end{bmatrix}\begin{bmatrix}y_1\\ y_2\end{bmatrix}$$

$$\det(A - \lambda I) = \det\left(\begin{bmatrix}4-\lambda & 1\\ 4 & 4-\lambda\end{bmatrix}\right) = (4-\lambda)^2 - 4$$

9) $(4-\lambda)^2-4=0$

$\lambda^2-8\lambda+12=0$

$(\lambda-6)(\lambda-2)=0$

$\lambda=6,2$

$P=\lambda_1+\lambda_2=8,\ Q=\lambda_1\lambda_2=12,\ \Delta=(\lambda_1-\lambda_2)^2=16$

$P>0, Q>0, \Delta\geq 0 \Rightarrow$ **Unstable Node**

$\lambda=6,\ (A-6I)x=0$

$\begin{bmatrix}-2 & 1\\ 4 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=0 \quad -2x_1+x_2=0 \quad x_2=2x_1 \Rightarrow x^{(1)}=\begin{bmatrix}1\\ 2\end{bmatrix}$

$\lambda=2,\ (A-2I)x=0$

$\begin{bmatrix}2 & 1\\ 4 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=0 \quad 2x_1+x_2=0 \quad x_2=-2x_1 \quad x^{(2)}=\begin{bmatrix}1\\ -2\end{bmatrix}$

$y=c_1\begin{bmatrix}1\\ 2\end{bmatrix}e^{6t}+c_2\begin{bmatrix}1\\ -2\end{bmatrix}e^{2t}$

$$\boxed{y_1=c_1e^{6t}+c_2e^{2t} \qquad y_2=2(c_1e^{6t}-c_2e^{2t})}$$